

A Behavioral Foundation for the Investment Wedge

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Introduction

- Challenge: Generating realistic comovement driven by the Investment Wedge
 - Among real variables and financial variables
- Behavioral firms systematically underestimate competitors' investment
 - Closely related to:
 - GREENWOOD & HANSON (2015), GH: "competition neglect" (CN)
 - ANGELETOS & LIAN (2023), AL: limited GE understanding
- Results:
 1. Solution method to introduce CN into GE model
 2. Endogenous Investment Wedge in RBC
Endogenous state (capital) implies boom-bust dynamics
 3. Right comovement among real and financial variables
Even in the absence of other frictions
 4. Return predictability, which match firm earnings survey evidence
(BORDALO, GENNAIOLI, LA PORTA, OBRIEN & SHLEIFER (2024))

1. Toy model: Muth (1961) with Competition Neglect
2. Real Business Cycle Model with Competition Neglect
 - Overview of Setup
 - Overview of Solution Method
 - Result: Analytic Expression for Endogenous wedge
3. Quantitative Analysis
 - Comovement: Real Variables
 - Comovement: Financial Variables
 - Matching Survey Data
4. Conclusion

Example: Muth (1961) with CN

"Rational Expectations and the Theory of Price Movements"

Muth, J., *Econometrica* 1961

An isolated market for a commodity:

- Demand: $Q_t^d = A_t - \beta P_t, \quad \beta > 0$

Demand Shifter: $A_t = \rho A_{t-1} + \epsilon_t, \epsilon_t \text{ i.i.d. } N(0, \sigma_\epsilon^2)$

- Supply: Time-to-build investment: $l_{it} = \gamma \mathbb{E}_{it}^\omega[P_{t+1}], \quad \gamma > 0$

Aggregate Investment: $I_t = \int l_{it} di$

Supply: $Q_{t+1}^s = I_t + (1 - \delta)Q_t$

- Market Clearing: $Q_t^d = Q_t^s = Q_t$

Competition Neglect Assumption

- Each firm i perceives aggregate investment as:

$$\mathbb{E}_{it}^{\omega}[I_t] = \omega I_{it}, \quad 0 \leq \omega \leq 1$$

- $\omega = 1$: rational expectations
 - $\omega < 1$: **competition neglect** (underweights others' responses)
-
- ω is the CN parameter
 - Failure to understand market equilibrium
 - Under-estimate reaction of aggregate investment
 $\implies$ Firms end up under-estimating future Q_{t+1}
Over-estimate future price P_{t+1}
 - $\in$ family of theories covered in [ANGELETOS & LIAN 2023](#), in a NK consumption model

Solution

- Use market clearing + invert demand function, plug into individual investment:

$$I_{it} = \frac{\gamma}{\beta} \mathbb{E}_{it}^{\omega} \underbrace{[A_{t+1} - Q_{t+1}]}_{P_{t+1}}$$

- Use CN assumption + capital dynamics:

$$I_{it} = \frac{\gamma}{\beta} [\rho A_t - (\omega I_{it} + (1 - \delta) Q_t)]$$

- **RED**: Source of boom, at t , and onward (over-investment)
ORANGE: Source of bust at $t + 1$, and onward (stock overhang correction)

$$I_{it} = \frac{\gamma}{\beta + \gamma \omega} (\rho A_t - (1 - \delta) Q_t)$$

A tug-of-war between these two effects on investment

- Boom, similar to [ANGELETOS & LIAN \(2023\)](#)

Proposition (Boom)

Assume that $\delta = 1$. Furthermore, assume that at $t - 1$ the economy is in steady state, and $\epsilon_t > 0$. Then, the IRF of investment is such that:

$$I_{t+\tau}^{CN} > I_{t+\tau}^{RE} > 0, \quad \forall \tau > 0$$

- CN generates over-investment, with decaying dynamics at rate ρ
- Only effect in **RED** is present

Boom-Bust With Endogenous State

- Endogenous state with partial depreciation brings novelty

Proposition (Boom-Bust)

Assume that $\delta < 1$. Assume that at $t - 1$ the economy is in steady state, and $\epsilon_t > 0$. Then, there is a $\rho(\omega)$ and a $t^* > t$ such that, for $\rho \in [\rho(\omega), 1 - \delta)$:

- Initially, the IRF of investment features a boom:

$$I_{\tau}^{CN} > I_{\tau}^{RE} > 0, \quad \forall \tau < t^*$$

- Later, the IRF of investment features a bust:

$$I_{\tau}^{CN} < 0 < I_{\tau}^{RE}, \quad \forall \tau \geq t^*$$

- Boom at t , investment crosses zero once, converges to zero from below
- Tug-of-war between RED and ORANGE, modulated by ρ and δ

Real Business Cycle (RBC) Model

Brief Setup Description, and Results

- Embed into canonical Real Business Cycle:
 - Households, firms, government
 - Maximizing agents
 - Competitive goods and labor markets
 - Economy buffeted by shocks to Total Factor Productivity (TFP)
- Novelty: Behavioral firms in capital-good producing sector
 - Adjustment costs, Q theory
 - Follow GH closely for specification of CN bias
- **No other frictions!**

- Standard RBC model
 - separable preferences (CRRA + power in labor), Cobb-Douglas technology, competitive final goods producers with flexible prices, competitive labor market
- Persistent AR(1) shocks to Total Factor Productivity (TFP)
 - **Single shock**: investment wedge is endogenous
 - Compare endogenous wedge to exogenous wedge
 - Shocks to marginal efficiency of investment (GHH, JPT)
- Notice that in GE:
 1. interest rate is endogenous
 2. rental rate functional form is determined by the production function(See paper for a discussion of technical issues dealt with by solution method)

- Loglinearize around steady state
- Roadmap:
 1. Do business cycle accounting
Find: CN observationally equivalent to investment wedge
 2. Calibrate model and simulate
Find: Match comovement financial aggregates
 3. Introduce corporate finance elements
Find: Match LTG earnings per share survey evidence

Business Cycle Accounting: CN is observationally equivalent to an investment wedge:

$$\tau_{lt} = \nu_k \hat{k}_t + \nu_a \hat{a}_t$$

Key property: ARMA(2,1) representation

$$\tau_{lt} = (1 + \psi_{lk} + \rho_a)\tau_{lt-1} - (1 + \psi_{lk})\rho_a\tau_{lt-2} + \text{MA terms}$$

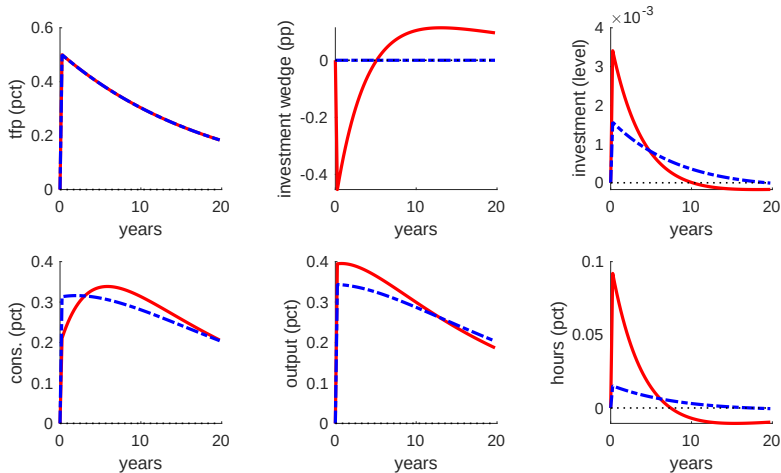
Why this matters:

- Dynamics of τ_{lt-1} and τ_{lt-2} produce **boom-bust** investment
- Wedge acts like **subsidy** initially, then **tax** later (tug-of-war)
- Endogenous state (capital) interacts with behavioral friction

Calibration

Parameter		Value
β	Time preference	0.99
γ	Inverse EIS	1
χ	Labor disutility	1.0324
φ	Inverse Frisch elasticity	1
μ	Mean growth rate	0.02/4
δ	Capital depreciation	0.025
α	Capital share	1/3
ϕ	Adjustment costs	20
ρ_a	TFP persistence	$0.95^{1/4}$
σ_a	TFP shock std dev	0.005
ω	CN parameter	0.1

IRFs: Real Variables



Red (solid): with competition neglect. Blue (dashed): frictionless model.
Deviations from balanced-growth path.

Stock value and return:

$$V_t = D_t + \mathbb{E}_t M_{t+1|t} V_{t+1}, \quad R_{st+1} = \frac{V_{t+1}}{V_t - D_t}$$

Mechanism:

- Firms equate *perceived* expected return to risk-free rate
- But perceptions are distorted $\implies$ actual equity premia $\neq 0$ (econometrician's)

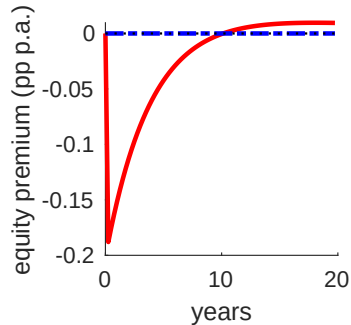
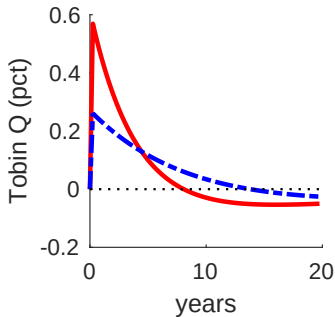
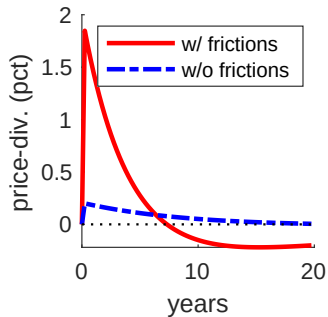
Key result: Investment wedge $\iff$ excess returns

$$\mathbb{E}_t[r_{st+1} - r_t] = \tau_{lt} - \beta \mathbb{E}_t[\tau_{lt+1}]$$

Properties:

- High valuations predict *low* future excess returns
- Matches Campbell-Shiller decomposition ($\sim 80\%$ return predictability)

IRFs: Financial Variables

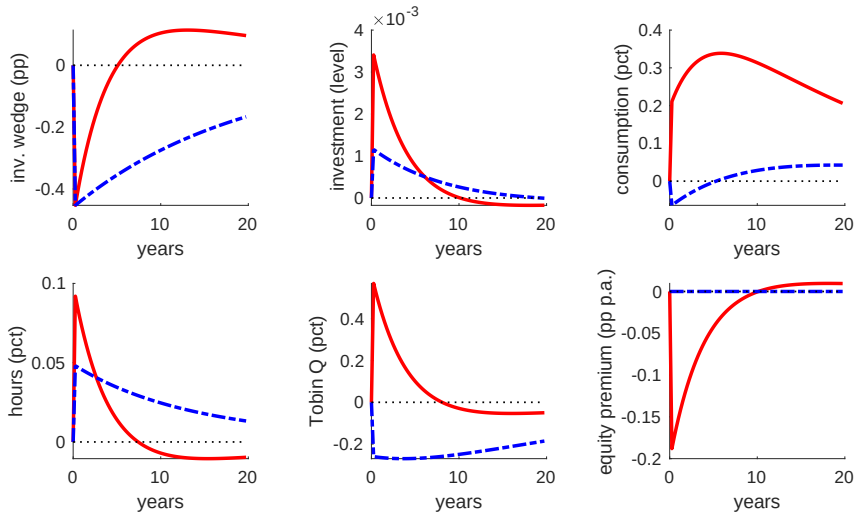


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Why Not Just Use Exogenous Investment Shocks? (Justiniano et al. 2010)

Visual Comparison: Same Wedge on Impact, Different Dynamics

IRF to a +1-s.d. shock, w/o friction, devs from balanced-growth path



Summing Up: CN vs Exogenous Investment Shocks

Why not just use exogenous investment shocks? (Justiniano et al. 2010)

	Exogenous Wedge	CN (Endogenous)
Wedge dynamics	AR(1), one-signed	ARMA(2,1), switches sign
Investment path	Persistently high	Boom <i>then</i> bust
Investment-Consumption comovement	Negative (comovement problem)	Positive ✓ (realistic)
Investment-Tobin's Q comovement	Negative (<i>wrong sign!</i>)	Positive ✓ (as in data)
Excess return predictability	Zero (no EP movement)	Time-varying ✓ (high $V \Rightarrow$ low $\mathbb{E}[R]$)

Empirical exercise:

$$FE_{t,t+h} = B_h \times LTG_t + \varepsilon_{t,t+h}$$

h (Q)	0	1	2	3	4	5	6	7	8	9	10
STG ₁	9.99	12.58	13.82	13.80	13.21	12.25	11.15	9.67	7.47	5.26	3.35
LTG	3.69	3.49	3.04	2.38	1.53	0.58	-0.33	-1.14	-1.63	-1.81	-1.69

Economic interpretation: Analysts FE exhibit boom-bust beliefs

How Can Model Match Volatility and Regression Coefficients? Adding Corporate Finance Features

Leverage ratio = 0.50:

- Affects volatility of stock returns
- Consistent with empirical leverage ratios

Dividend smoothing (λ_{div}):

- Lintner-style smoothing: $D_t = \lambda D_{t-1} + (1 - \lambda) \text{FCF}_t$
- Affects dividend dynamics
- Under Modigliani-Miller, doesn't affect real allocation

Result: Corporate finance features help match volatility moments without affecting core mechanism

Can Match:

1. **LTG volatility:** Close to target 2.20%
2. **LTG Coefficients:** Positive coefficients at short horizon
3. **Reversal:** Declining regression coefficients

Main insight: Competition neglect can account for forecast error patterns

Methodological Contribution

General solution method for competition neglect in nonlinear GE models

Economic Contribution

A *single* behavioral friction (CN) in the investment block:

- Microfounds the Investment Wedge; generates boom-bust dynamics
- Produces realistic macro *and* financial comovement
- Without adding financial, search or nominal frictions

Key Takeaway: A disciplined expectation bias in GE generates *both* macro boom-bust *and* asset-pricing patterns that exogenous shocks cannot replicate.